

Numerical Methods on Global Monthly SST Means

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1. Introduction

The focus of this report is changes in sea surface temperature (SST) in relation to climate change. SST plays a vital role in regulating global climate systems, and fluctuations can have wide-ranging impacts, including shifting storms, prolonged droughts, and increased bacteria growth seasons (Environmental Protection Agency, 2025).

SST is affected by a myriad of variables, with time of year and geographic location being two primary factors. Others also include precipitation, atmospheric oscillations, and solar radiation. Additionally, sea surface temperature fluctuations are correlated with some natural phenomena, such as El Niño-Southern Oscillation (ENSO) patterns: positive SST abnormalities can indicate El Niño phases and negative anomalies can indicate La Niña phases (World Meteorological Organization, n.d.). ENSO patterns are divided by region. In particular, the Niño 3.4 region lies between 5°N-5°S and 170°W-120°W. Anomalies in this region are considered to represent "the average equatorial SSTs across the Pacific from about the dateline to the South American coast," (Trenberth, 2025). When SSTs in this region exceed $|0.4^{\circ}\text{C}|$ for a period of six or more months, it is considered an El Niño or La Niña occurrence (Trenberth, 2025).

The data used in this report are from the COBE-SST 2 dataset containing monthly SST means from January 1850 to March 2025, measured on a 1.0° latitude by 1.0° longitude global grid (89.5°N - 89.5°S, 0.5°E - 359.5°E). The dataset was collected by the Japan Meteorological Agency (JMA) to analyze the relationship between sea surface temperatures and ice concentrations. Before the 1980s, data was collected via ships and buoys. Satellite observations were introduced in the 1980s and helped fill in the gaps between the ship and buoy data (Japan Meteorological Agency, 2025).

This report aims to analyze long-term trends in monthly SST means to gain understanding as to how they may continue to fluctuate in the future. It will also analyze anomalies to identify ENSO patterns and extrapolate for future phases.

2. Methods and results

2.1. Data

The dataset was acquired from the NOAA Physical Sciences Laboratory. It contains SST data in NetCDF4 format, with latitude, longitude, and time as coordinates. To maintain multidimensionality, the data were imported as an xarray, which was primarily used to analyze long-term global trends with plots showing coordinate-specific results. The dataset was also converted into comma-separated values (CSV) format to identify abnormalities and extrapolate for potential future ENSO phases with plots showing generalized results over time. The CSV dataset is a subset of the original, focusing on SST from 1988 to 2025. This time range was chosen because by 1988, blended SST analysis had become widely distributed (Reynolds and Smith, 1994), meaning that both in situ and satellite collection methods employed which resulted in more accurate data from 1988 to present.

2.2. Smoothing

To identify trends in SST changes, the xarray data were smoothed temporally and spatially to minimize noise. A three-month rolling mean was first applied to temporally smooth short-term fluctuations. Then cubic spline was applied to temporally smooth long-term fluctuations, demonstrating gradual sea surface temperature changes over time. Cubic spline was also used to interpolate missing data which were flagged in the dataset with value -9.96921e+36f. Lastly, a Gaussian filter smoothed the data spatially to focus on regional

patterns. To minimize filtering and blurring near coastlines, a sigma of 0.3 was used near coasts versus a sigma of 1.0 over the ocean.

2.3. Linear Regression

After smoothing, linear regression was applied to the time series at each grid point. The regression calculated the linear slope ($^{\circ}\text{C}/\text{year}$) and the p-value at each grid point. The same linear regression function was applied to both the entire dataset from 1850 to 2025 and a localized subset of the data from 1988 to 2025, chosen because of the accuracy of the data.

Fig. 1 shows the calculated rate of change at each grid point for the two datasets. The 1850-2025 dataset shows that the majority of global sea surface temperatures are increasing at approximately $0.005^{\circ}\text{C}/\text{year}$; however, the more recent data (1988-2025) shows that temperatures may be rising more quickly as of late, with the majority of the world's SST rising between 0.02 and $0.04^{\circ}\text{C}/\text{year}$. The data from 1988-2025 also shows more instances of cooling.

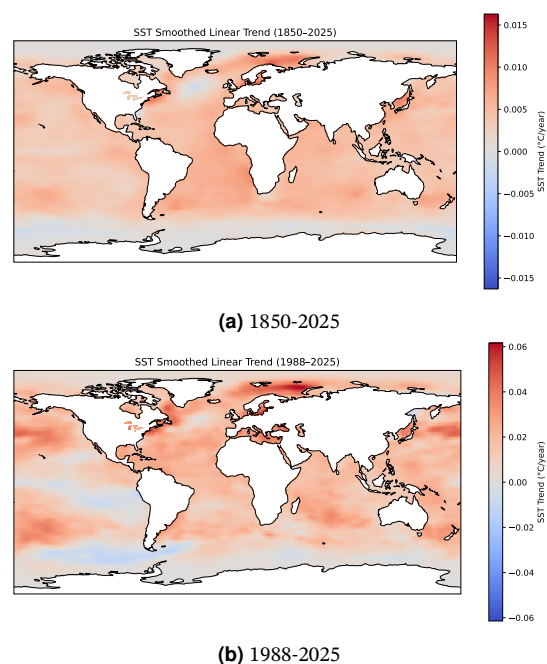


Figure 1. SST trends from linear regression analysis.

Fig. 2 shows the statistical significance of the linear regression: insignificant changes in SST have been removed and are shown in white. From the differences between Fig. 1 and Fig. 2, it can be seen that the majority of warming is statistically significant in both datasets. The cooling seen from 1988-2025 is not significant.

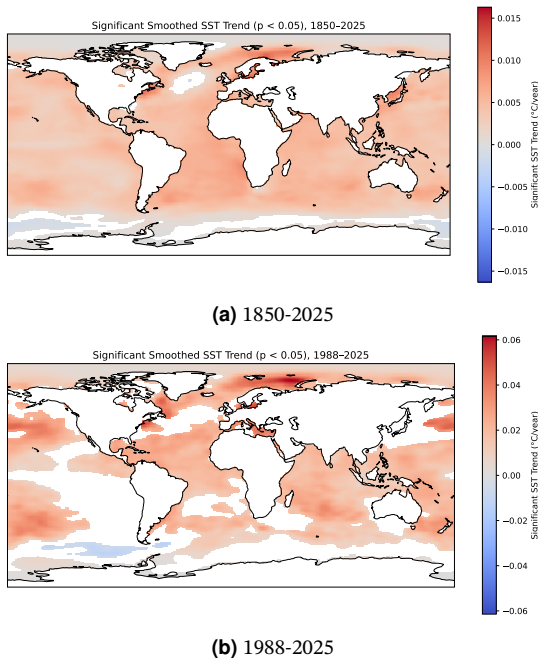


Figure 2. Statistically significant SST trends from linear regression analysis

These results were corroborated by Fig. 3, a plot of the yearly average SSTs created to visualize the changes in SST in linear form. To account for the latitude and longitude coordinates in the xarray, weights were applied to each latitude. The monthly SST means were recomputed given these weights and then the data was resampled by year and smoothed using a 10-year rolling mean.

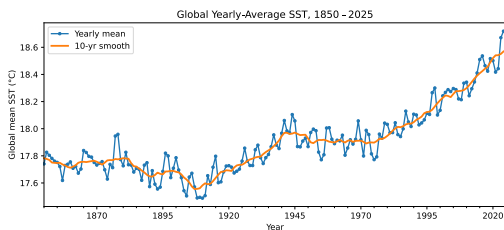


Figure 3. Global Yearly Average SST (1850-2025)

Based on the plot, the primary historical warming periods occurred between approximately 1910-1945 and 1970-present. The only cooling period occurred between 1880-1905.

Given significant results of the linear regression analysis and the global yearly average SST plot, it is clear that global sea surface temperature is on the rise.

2.4. ENSO Phase Detection

Fig. 4 is a representation of the six-month Niño anomalies from 1988 to 2025, with El Niño and La Niña's episodes shaded in red and blue, respectively. Months were classified as warm, cold, or neutral depending on if the smoothed anomaly exceeded $|0.4^\circ\text{C}|$ or not, and only sequences lasting at least 6 months (non-neutral) were then classified as El Niño or La Niña.

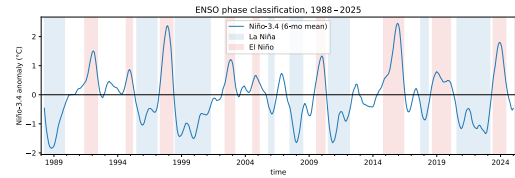


Figure 4. ENSO Phase Classification (1988-2025)

The 1997-1998 and 2014-2016 El Niño events reached peaks higher than $+2^\circ\text{C}$, and the 2020-2022 La Niña episode persisted for more than two years. The swings we observe here are indicative of interannual variability, as individual ENSO phases can temporarily cause the SST to raise or lower by more than two degrees, much larger than the century-scale warming rate that was shown in the last section.

The dates that end each phase by crossing zero were verified with three root-finding methods, bisection, Newton-Raphson, and cubic spline segmentation (Fig. 5). All methods agreed on these dates to within one day, confirming that the phase identification is insensitive to the root-finding technique applied.

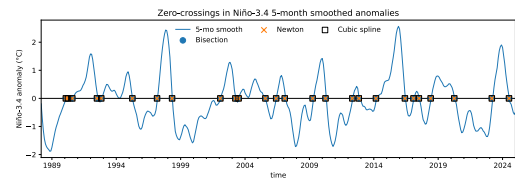


Figure 5. Root Finding in Niño 3.4 Anomalies

ENSO acts as a rapidly changing modulation on an otherwise rising baseline. Because of this, accounting for El Niño and La Niña's episodes is important when looking at short-term SST trends as they may distract from the larger overall trends of steadily rising temperatures. The earlier smoothing methods help prevent ENSO phases from distracting from the long-term trends under analysis.

To show how this might play out under a purely statistical continuation of how it has been, Fig. 6 uses a seasonal ARIMA model to simulate 20 additional years of variability with the future predictions starting at the vertical grey line. This simulated extension produces a familiar alternation of phase episodes with similar peaks. Obviously, this model considers only statistical guesses about the future and does not contain information about the general trend of the climate, the effect of greenhouse gases, or any other factors.

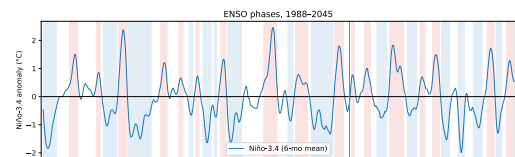


Figure 6. ENSO Phases Extrapolated (1988-2045)

While these phases show a variability year-to-year in SSTs, the overall trend of the climate (Fig. 3) is nonetheless on a steady rise. It is important to note that focusing on the ENSO phases are not an effective way to communicate the state of the global climate.

3. Conclusion

3.1. Global SST

Since measurements began in 1850, the global sea surface temperature has risen at an average of about 0.005°C per year; however, in the latter half of the century this number has jumped significantly to about 0.02 to 0.04°C per year. Some of this jump may be attributed to

the increased accuracy that accompanied the introduction of satellite observations in the 1980s, but most of it is related to the post-industrial period's boom in greenhouse gases. Using numerical techniques, the data was smoothed to show that the warmer parts of the ocean are getting warmer and the few spots that could be cooling are not statistically significant.

Rising SST will have significant impacts to both marine life and global climate systems. They will threaten ocean life such as corals, increase frequency of harmful algal blooms, and reduce ocean nutrient supply. This will dramatically affect marine ecosystems, leading to potential declines in fish populations, (Environmental Protection Agency, 2025). Atmospherically, increases in SST will lead to increased water vapor over the oceans, resulting in heavier rains and snow and potentially leading to more frequent and more harmful disasters such as hurricanes and typhoons, (Environmental Protection Agency, 2025).

The global increase in sea surface temperature is one instance of global warming. Global warming refers to the “long-term heating of Earth's surface observed since the pre-industrial period (1850-1900) due to human activities”, (NASA Science, 2024). While land areas are warming faster than oceans, it cannot be denied that oceans temperatures are increasing at an alarming rate. The European Geosciences Union published a report in 2016 stating that a 0.5°C global temperature increase would mean that heat waves would last around a third longer, rain storms would be approximately a third more intense, and sea levels would approximately one third higher, (NASA Science, 2016).

3.2. ENSO Phases

The analysis identifies ENSO phases, with the El Niños generally more extreme than the La Niñas. While these phases can give insights into annual means, they seemingly do nothing to derail the overall upward trend of the SSTs. The ARIMA forecast seems to suggest that the current El Niño phases will continue for a few more years until approximately 2029.

3.3. Limitations

This analysis is not without its limitations. As previously mentioned, measurements before the 1980s were taken by instruments on shorelines, ships, and buoys. In addition to simply being less accurate, these measurement limitations led to a massive undersampling of the ocean; most of the world's ocean is not directly along a 19th century trade route (NASA, 2020). Additionally, some of the smoothing techniques may have over smoothed the data. For example, Gaussian filtering required adaptive filtering along coastlines to avoid over blurring and distorting land areas. This may have decreased the accuracy of the data.

In terms of the ENSO phase projections, the ARIMA forecast is not a crystal ball; it is somewhat representative of a coin toss. The phase predictions may not be accurate, especially as time increases. ENSO phases help account for abnormalities within the data and should be smoothed when considering long-term projections.

3.4. Future enhancements

Future research could build on this report by tightening the effectiveness of our numerical methods and viewing the data in new ways. For example, by downscaling the scope to focus on specific regions and how they affect each other instead of on global averages, research could make the data actionable region-by-region. By using Bayesian statistical methods or by running a bootstrap on the data, researchers can gain confidence in their results. Folding in data from other similar sources such as the Multivariable ENSO Index (MEI) and the Southern-Oscillation Index (SOI), or from more general climate data such as atmospheric CO₂ concentration across the globe, polar ice cap volume, or glacier length changes, more specific conclusions can be made. Correlating ENSO patterns with real-world

catastrophic weather events, combined with machine learning tools like an attention-based neural network, could allow data scientists to pick out the more subtle warning signs of incoming climate, potentially allowing for more early detections of incoming heat waves or floods. And of course, collaboration with experts in the field would allow for a much greater understanding of the results and relationships between variables.

3.5. Closing Statements

Wobbles such as El Niño or La Niña occurrences come and go, but the ocean's thermostat is stuck on warmer. The regions that are cooler are statistically insignificant. Keeping track of both the year-over-year data and the big picture helps cut through the noise and keeps the climate conversation going. Keeping the climate conversation grounded in data analysis, and by extension continuing to fund every form of climate data collection and research, is essential to communicating what is really happening. As the acceleration of the mean trend continues, keeping up will require continued high-resolution observations and more rigorous, multi-method analysis.

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Video Link

<https://youtu.be/jXoJMuoDzeI>